

LABSHEET - 2**HIMANI CHAUDHARY****BTECH(CSE) 3RD YEAR****CU240250967****SECTION - B****ROLL NO. - 24****Mastery in Data Analytics and Visualizations and Career Advancement (Technical)**

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
# 1. LOAD IMAGE
# =====

image_path = "/content/modern-business-building-with-glass-wall-from-empty-floor.jpg"

image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Check the image name and path.")
    print("Path used:")
    print(image_path)
    exit()
else:
    print("Image loaded successfully!")
    print("Himani Chaudhary")
    print("CU240250967")
```

Image loaded successfully!
Himani Chaudhary
CU240250967

```
# 2. CONVERT IMAGE TO GRAYSCALE
# =====

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 3. CONTRAST STRETCHING
# =====

min_pixel = np.min(gray)
max_pixel = np.max(gray)

if max_pixel == min_pixel:

    contrast_stretched = gray.copy()

else:

    contrast_stretched = (
        (gray - min_pixel) /
        (max_pixel - min_pixel) * 255
    ).astype(np.uint8)
    print("Himani Chaudhary")
    print("CU240250967")
```

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```
# 4. HISTOGRAM EQUALIZATION
# =====

hist_equalized = cv2.equalizeHist(gray)
```

```
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 5. CLAHE
# =====

clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray)
print("Himani Chaudhary")
print("CU240250967")
```

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```
# 6. DISPLAY ORIGINAL AND ENHANCED IMAGES
# =====

plt.figure(figsize=(15, 10))

# Original Grayscale
plt.subplot(2, 3, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale")
plt.axis("off")

# Contrast Stretching
plt.subplot(2, 3, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

# Histogram Equalization
plt.subplot(2, 3, 3)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

# CLAHE
plt.subplot(2, 3, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

Original Grayscale



Contrast Stretching



Histogram Equalization



CLAHE



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7. DISPLAY HISTOGRAMS

=====

```
plt.figure(figsize=(15, 8))
```

Original Histogram

```
plt.subplot(2, 2, 1)
plt.hist(gray.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
```

Contrast Stretching Histogram

```
plt.subplot(2, 2, 2)
plt.hist(contrast_stretched.ravel(), 256, [0, 256])
plt.title("Contrast Stretching Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
```

Histogram Equalization

```
plt.subplot(2, 2, 3)
plt.hist(hist_equalized.ravel(), 256, [0, 256])
plt.title("Histogram Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
```

CLAHE Histogram

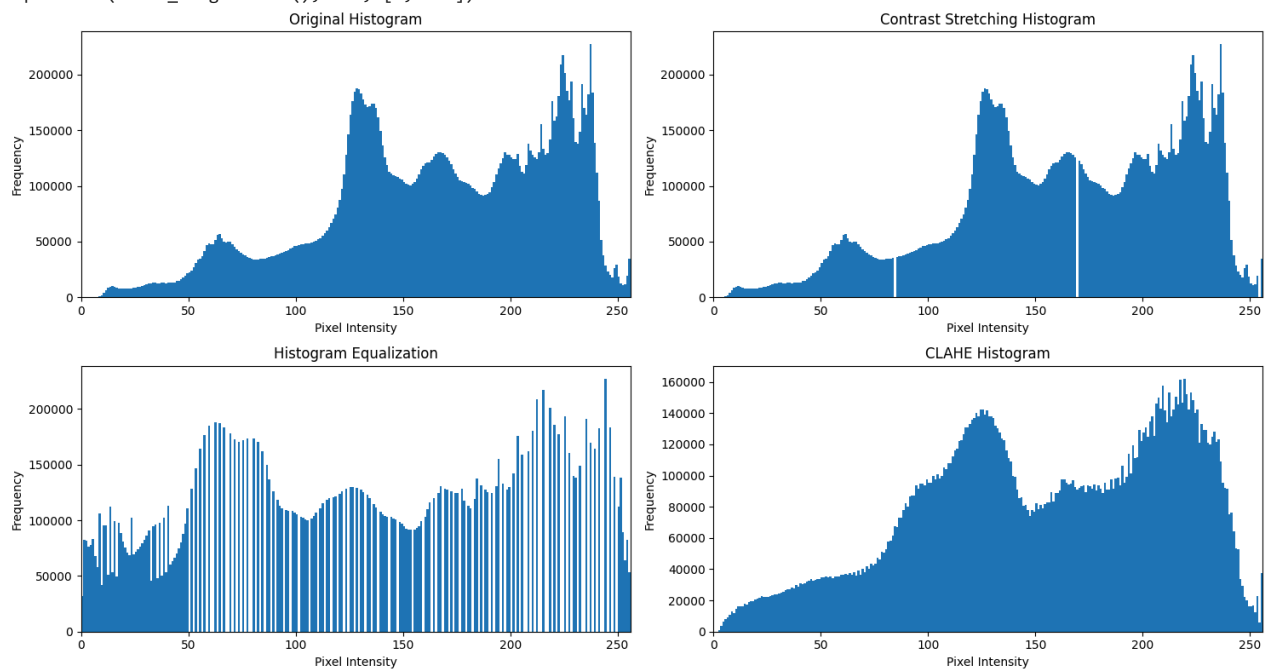
```
plt.subplot(2, 2, 4)
plt.hist(clahe_image.ravel(), 256, [0, 256])
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.xlim([0, 256])
```

```
plt.tight_layout()
plt.show()
print("Himani Chaudhary")
print("CU240250967")
```

```

/tmp/ipykernel_901/2896965051.py:9: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is depr
plt.hist(gray.ravel(), 256, [0, 256])
/tmp/ipykernel_901/2896965051.py:18: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is dep
plt.hist(contrast_stretched.ravel(), 256, [0, 256])
/tmp/ipykernel_901/2896965051.py:27: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is dep
plt.hist(hist_equalized.ravel(), 256, [0, 256])
/tmp/ipykernel_901/2896965051.py:36: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is dep
plt.hist(clahe_image.ravel(), 256, [0, 256])

```



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```

# 8. SAVE ENHANCED IMAGES
# =====

cv2.imwrite("contrast_stretched.jpg", contrast_stretched)
cv2.imwrite("histogram_equalized.jpg", hist_equalized)
cv2.imwrite("clahe_image.jpg", clahe_image)

print("=====")
print("Processing completed successfully!")
print("=====")
print("Saved files:")
print("1. contrast_stretched.jpg")
print("2. histogram_equalized.jpg")
print("3. clahe_image.jpg")
print("Himani Chaudhary")
print("CU240250967")

```

```

=====
Processing completed successfully!
=====
Saved files:
1. contrast_stretched.jpg
2. histogram_equalized.jpg
3. clahe_image.jpg
Himani Chaudhary
CU240250967

```